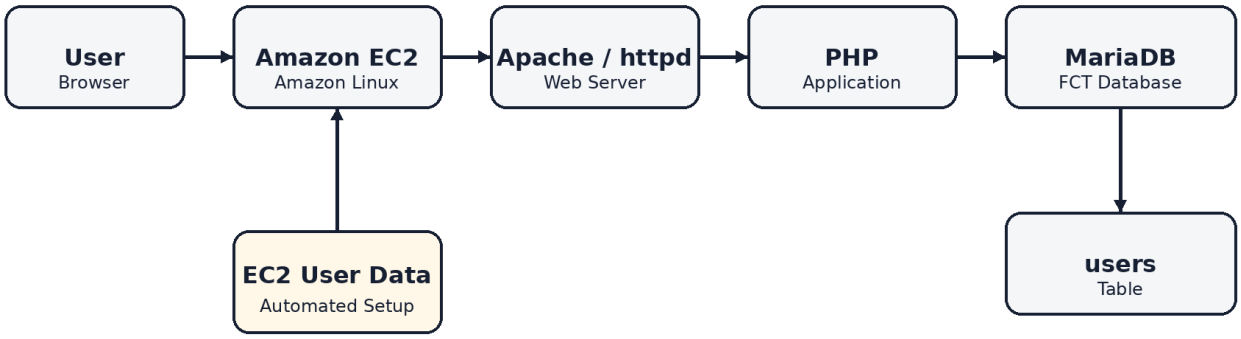
Dynamic PHP Website Hosting on Amazon Linux using LAMP

AWS EC2 • Apache • PHP • MariaDB • HTML

End-to-end project report with implementation flow, troubleshooting, verification and proof screenshots.

PHP Dynamic Website - LAMP Architecture



PROJECT STACK Linux + Apache/httpd + MariaDB + PHP

1. Project Overview

This project demonstrates how to host a dynamic PHP signup website on an Amazon Linux EC2 server using the LAMP stack. The application accepts signup information, processes it with PHP and stores the record in MariaDB.

2. Objective

Launch an EC2 server, automate the initial setup with User Data, configure Apache/PHP/MariaDB, deploy the application, connect PHP to MariaDB, troubleshoot the missing mysqli connector and verify the final database record.

3. AWS EC2 Configuration

Instance Name: **PHP Website**
Operating System: **Amazon Linux**
Instance Type: **t3.micro**
Key Pair: New key pair created during launch
Network: Existing security group selected

4. EC2 Instance Details

The EC2 instance was launched successfully. The highlighted Public IPv4 address was used to access the web server.

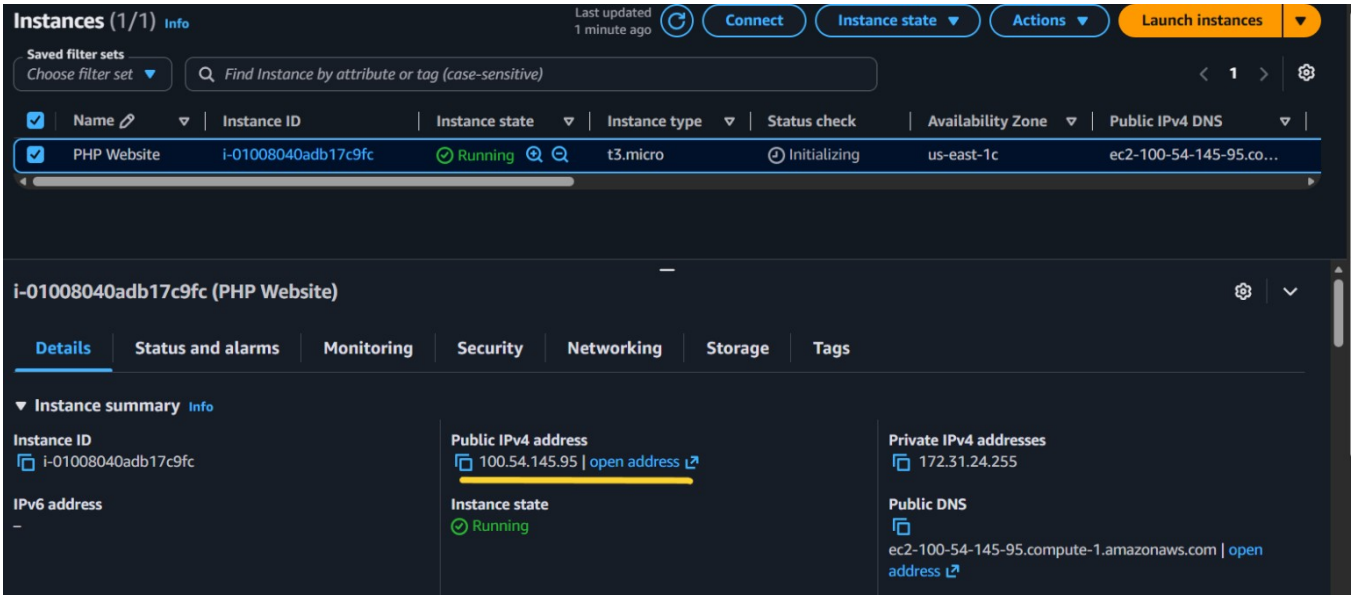


Figure 1 - EC2 instance and highlighted Public IPv4 address.

5. EC2 User Data Script

User Data automated package installation, service startup/enablement and initial web-file creation.

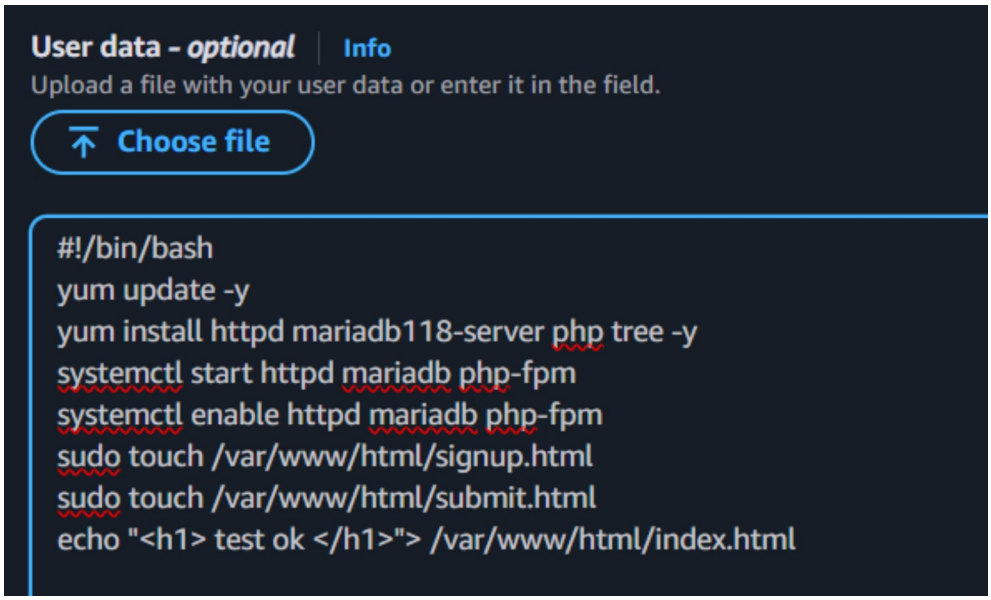


Figure 2 - EC2 User Data script used for initial server setup.

6. Initial Apache Verification

Opening the Public IPv4 address returned the Apache test page: **httpd test ok**.

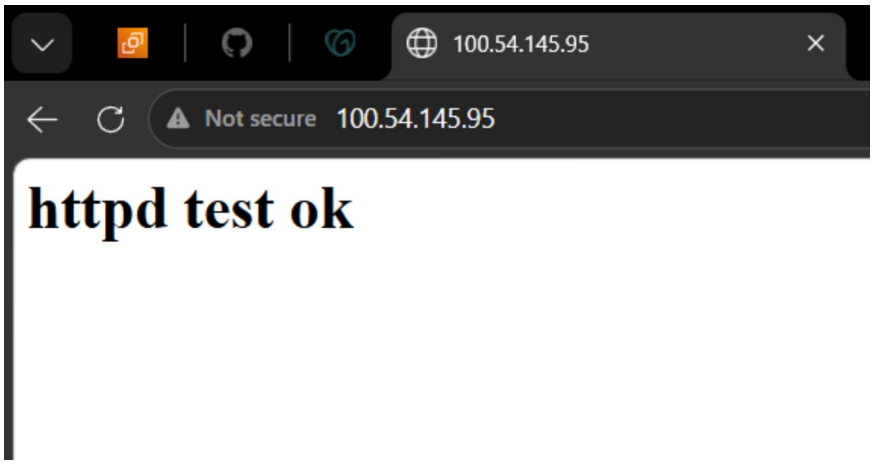


Figure 3 - Apache/httpd test result.

7. Service Verification

Apache, MariaDB and PHP-FPM were verified as **active (running)** using systemctl.

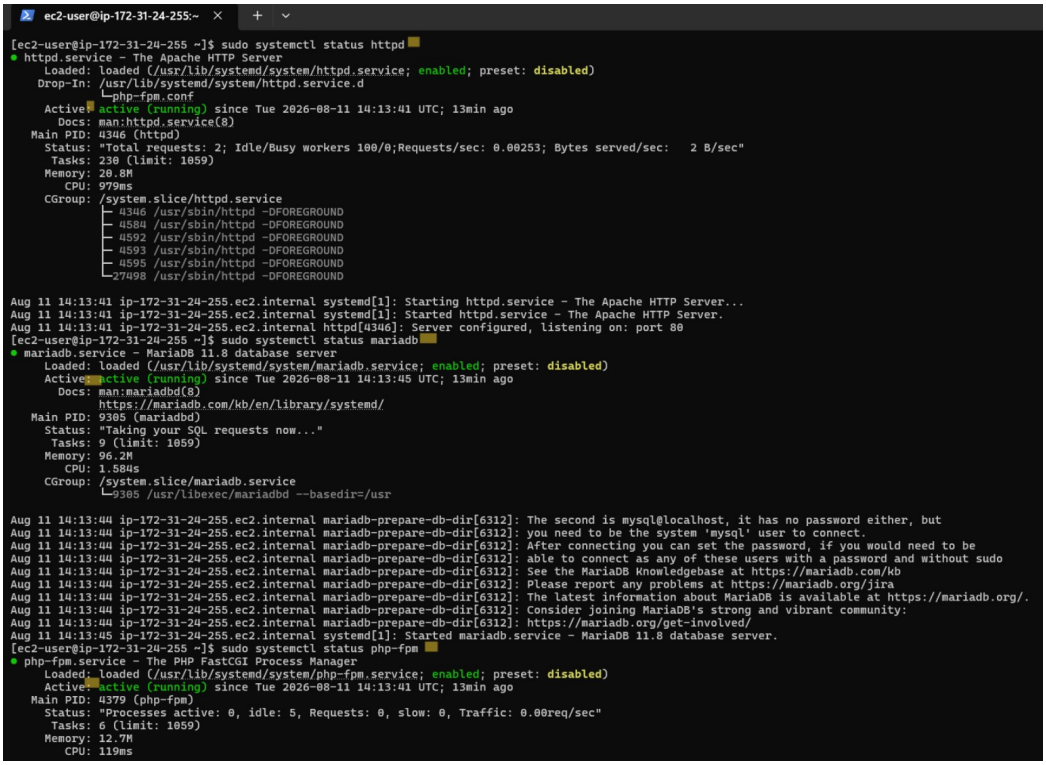


Figure 4 - httpd, MariaDB and PHP-FPM service verification.

8. Website Files

The application files **signup.html** and **submit.php** were verified under **/var/www/html/**.

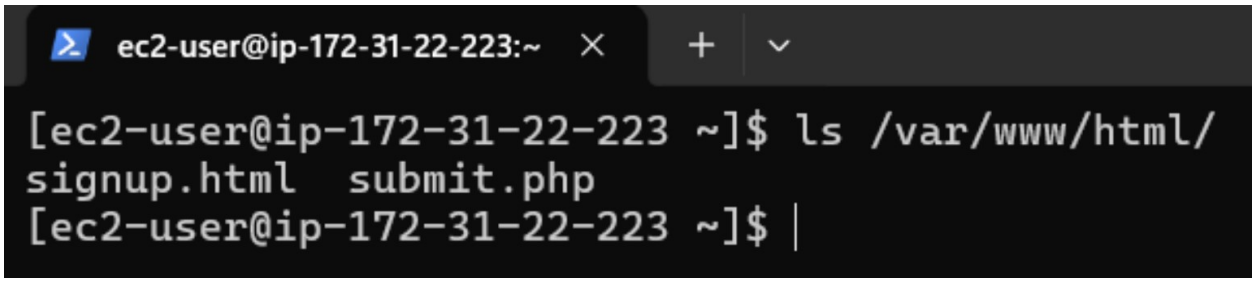


Figure 5 - Application files in the Apache web root.

9. MariaDB Database & Table Creation

MariaDB authentication was configured, the **FCT** database was created and the **users** table was defined with an auto-increment primary key.

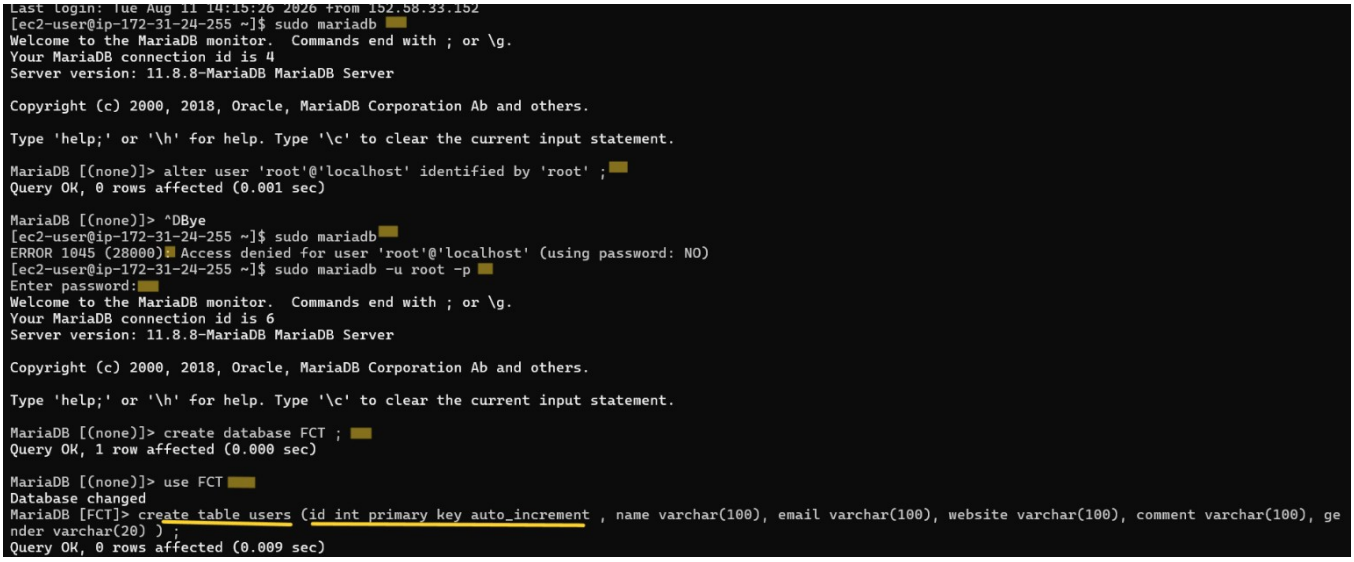


Figure 6 - MariaDB login, database creation and users table creation.

10. MySQL Connector Error

The first form submission failed because **mysqli_connect()** was unavailable.

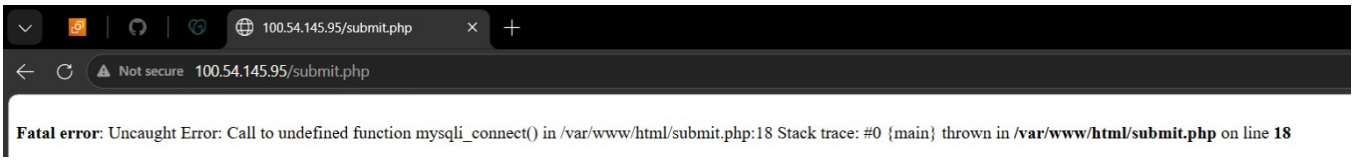


Figure 7 - PHP database connection error.

11. MySQL Connector Installation

PHP **8.5.9** was verified and the compatible **php8.5-mysqlnd** connector was installed.

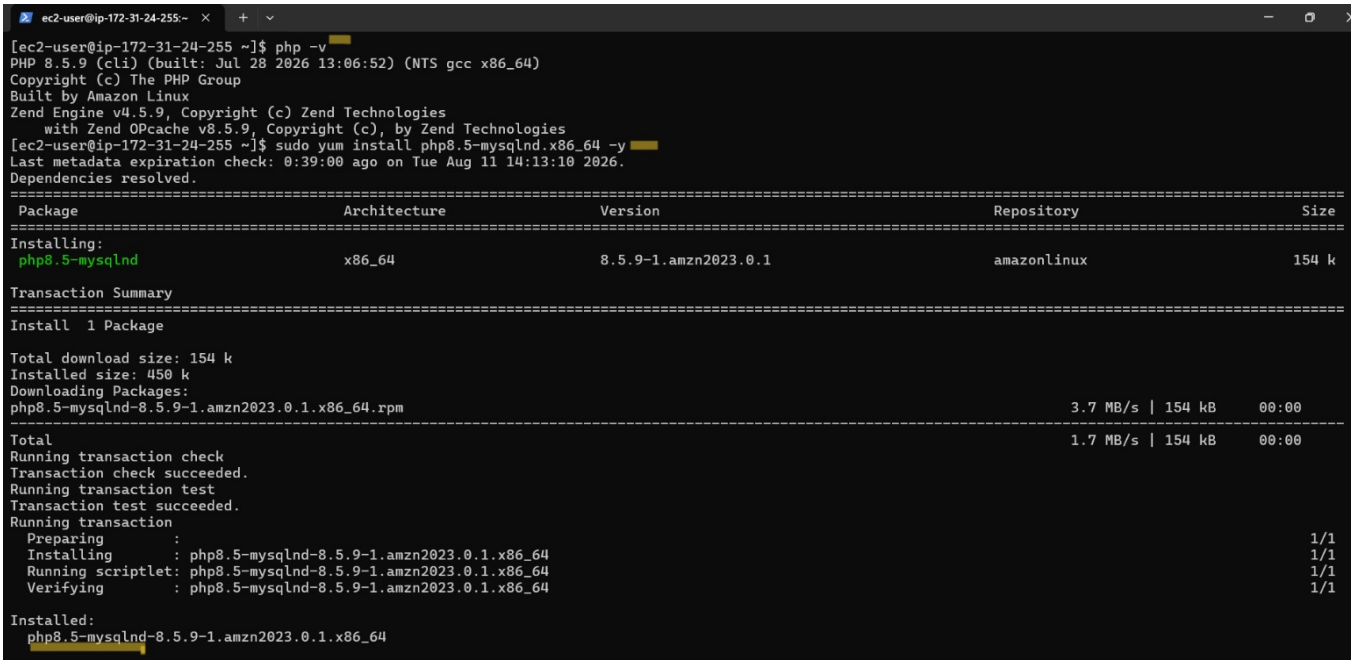


Figure 8 - PHP version check and compatible MySQL connector installation.

12. Signup Form Input

The deployed signup form was filled and prepared for submission.

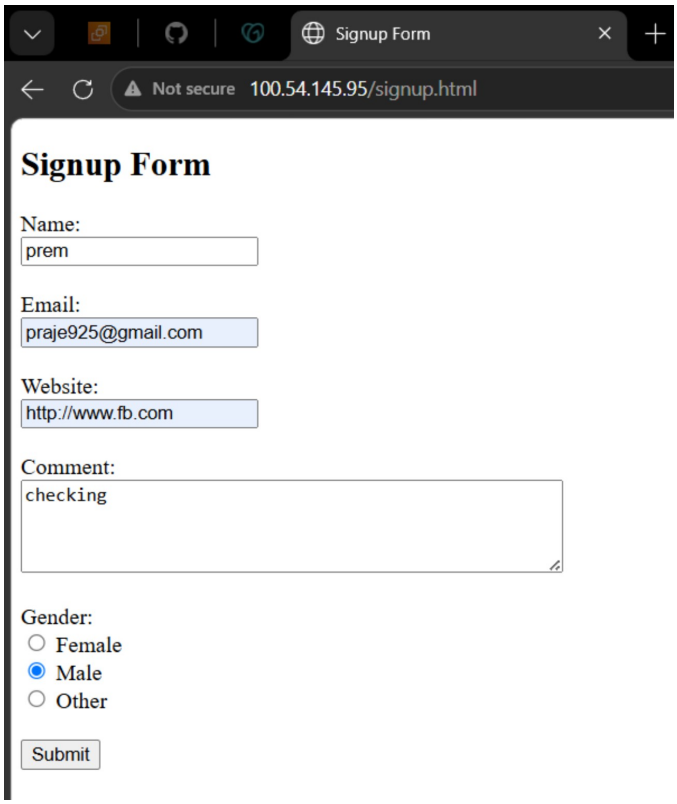


Figure 9 - Signup form input.

13. Form Submission Output

The PHP application successfully processed the form and reported that a new record was created.

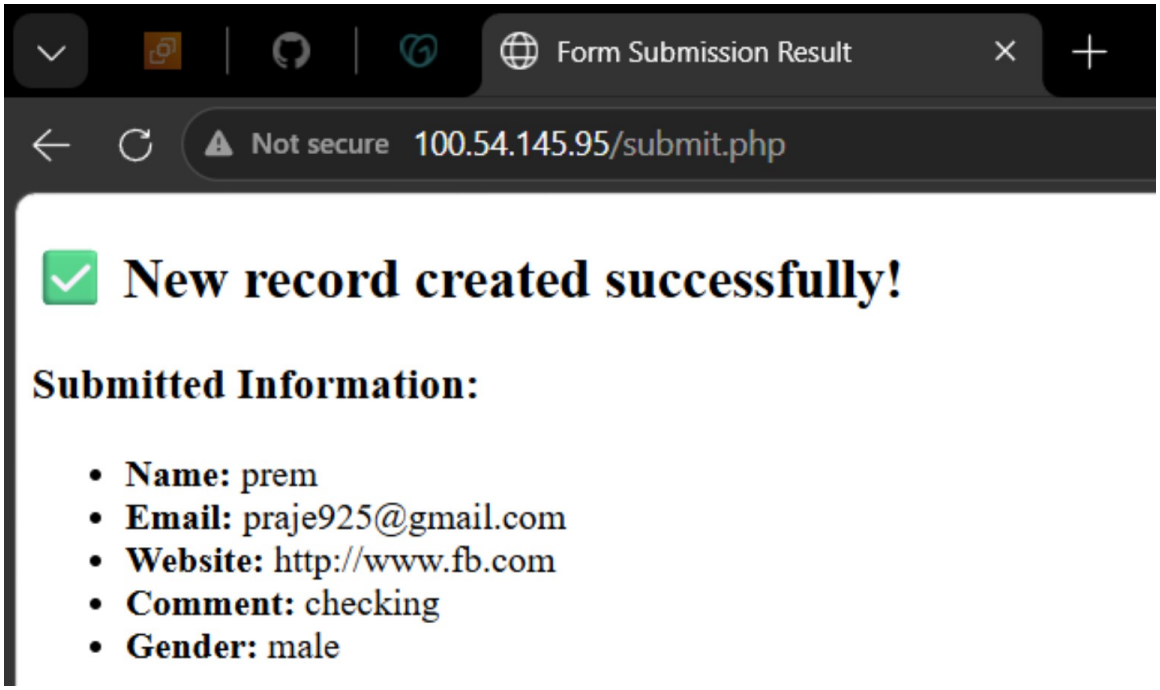


Figure 10 - Successful form submission response.

14. Database Record Verification

The inserted record was verified directly in the MariaDB **users** table using **SELECT * FROM users;**.

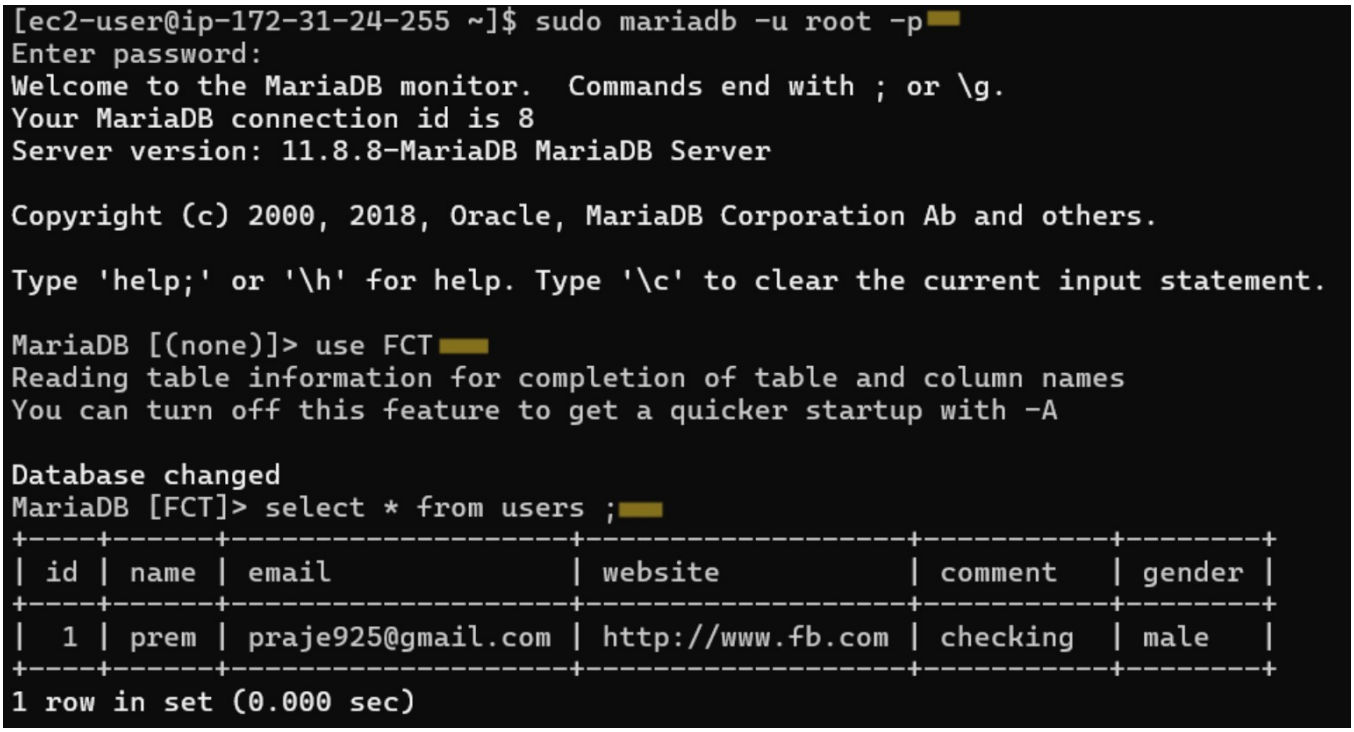


Figure 11 - Final database record verification.

15. Application Flow

Browser → signup.html → submit.php → PHP mysqli → MariaDB FCT → users → record stored successfully

Browser	signup.html	submit.php	PHP + mysqli	MariaDB	users

16. Problems Faced & Solutions

Problem: Call to undefined function mysqli_connect().
Reason: Required PHP MySQL connector was missing.
Solution: Checked PHP 8.5.9 and installed php8.5-mysqlnd.
Result: PHP successfully communicated with MariaDB.

17. What I Learned

I learned practical AWS EC2 deployment, User Data automation, Linux service verification, Apache hosting, PHP form handling, MariaDB database creation, PHP-MariaDB connectivity and basic troubleshooting.

18. Future Improvements

For a production version, I would use prepared statements, secure credential management, HTTPS, a dedicated database user and stronger input validation.

19. Conclusion

This project helped me understand how a PHP application can be hosted on AWS from start to finish. I learned how to launch an EC2 server, automate setup, configure Apache/PHP/MariaDB, create a database, solve a connector error and verify that form data was stored successfully.

FINAL RESULT

Dynamic PHP Signup Website Successfully Hosted on Amazon EC2